

MTH 446: Lecture # 9

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Lecture Span

1. Continuous complex functions

Continuous complex functions

Recall that the definition of a continuous complex function:

Definition 0.1. Let $D \subseteq \mathbb{C}$ and z_0 is an accumulation point of D and $f : D \rightarrow \mathbb{C}$. Then, the function f is continuous if

1. $z_0 \in D = \text{dom}(f)$
2. $\lim_{z \rightarrow z_0} f(z)$ exists
3. $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

The algebraic properties of continuous functions. Let $D \subseteq \mathbb{C}$ and z_0 be an accumulation point of D . Let $f, g : D \rightarrow \mathbb{C}$ be continuous functions at $z = z_0$. Let $c \in \mathbb{C}$, then

1. (cf) is continuous at $z = z_0$
2. $f + g$ is continuous at $z = z_0$
3. fg is ...
4. If $g(z_0) \neq 0$, then f/g is ..., in particular, $1/g$ is ...

Example

Let $n \in \mathbb{N}$ and $P_n(z) = a_0 + a_1z + a_2z^2 + \dots$ where $a_n \neq 0$ and $a_j \in \mathbb{C}$ for all j . Then, each of the monomials are continuous in $\mathbb{C}$.

Remark: suppose $D \subseteq \mathbb{C}$, then f, g cont. implies $\alpha f + \beta g$ also cont.

Also, fg is also cont.

Infinite limits of complex functions

Recall, $\lim_{z \rightarrow \infty} f(z) = w \iff \forall \epsilon > 0, \exists \delta > 0 : 0 < |z - z_0| < \delta, z \in \text{dom}(f) \implies |f(z) - w_0| < \epsilon$ (definition of limit).

To every ϵ -neighborhood of w_0 , corresponds a deleted δ neighborhood of z_0 such that

$$f(\overset{\circ}{D}_\delta(z_0) \cap D) \subseteq D_\epsilon(w_0) \quad (1)$$

Now, introduce the infinite point ∞ . Define the Riemann sphere. The sphere is an ordinary unit sphere S^2 in $\mathbb{R}^3$ given by

$$x_1^2 + x_2^2 + x_3^2 = 1 \quad (2)$$

We project the complex plane onto the R sphere. This is also called the stereographic projection. The infinite points of the complex plane are mapped onto the top of the R sphere, also the north pole N .

Consider the point $z = x + iy$. Draw a line from the point z to the North Pole, and this line intersects at a single point on the sphere (this is where z gets mapped onto the sphere).

Note, only the infinite points of the complex plane get mapped directly onto N .

Therefore, $\mathbb{C} \cong S^2/\{N\}$, that is the complex plane is isomorphic to this Riemann sphere minus the north pole. N represents infinity.

Definition 0.2. The extended complex plane is $\cup \{\infty\}$. This is isomorphic to the entire R sphere.

Definition 0.3. The distance between two points of the R sphere is the standard euclidean distance in $\mathbb{R}^3$.

Definition of limit at infinity. Then, $\overset{\circ}{D}_\epsilon(\infty)$ is defined as

$$\overset{\circ}{D}_\epsilon(\infty) = \{z \in \mathbb{C} : |z| > 1/\epsilon\} \quad (3)$$

$1/\epsilon$ is the radius of the projected circle onto the x-y plane of the deleted epsilon neighborhood.

Definition 0.4. Let $G \subseteq \mathbb{C}$ and $z_0 \in \mathbb{C}$ be an accumulation point of G . Let $f : G \rightarrow \mathbb{C}$. Define

$$f(\overset{\circ}{D}_\delta(z_0) \cap G) \subseteq D_\epsilon(\infty) \subseteq \overline{\mathbb{C}} \quad (4)$$

Therefore,

$$f(\overset{\circ}{D}_\delta(z_0) \cap G) \subseteq \mathbb{C} \quad (5)$$

$$f(D_\delta(z_0) \cap G) \subseteq \overset{\circ}{D}_\delta(z_0) \quad (6)$$

Therefore, $\lim f(z) = \infty$ iff for all deleted ϵ neighborhood at ∞ exists deleted δ D of z_0 such that

$$f(\overset{\circ}{D}_\delta(z_0) \cap G) \subseteq \overset{\circ}{D}_\epsilon(\infty) \quad (7)$$

Let z_0 be an accumulation point of G , then deleted ϵ neighborhood intersect G .

If $z_0 = \infty$, then deleted ϵ neighborhood intersects G .

If $z_0 \in \mathbb{C}$, then

$$\lim f = \infty \iff 0 < |z - z_0| < \delta, z \in G \implies |f(z)| > 1/\epsilon.$$

$$\lim f(z) = \infty, \text{ such that } |z| > 1/\delta \text{ and } |f(z)| > 1/\epsilon.$$

For $z \rightarrow \infty$, use Riemann sphere to let $z > 1/\delta$.

Useful fact: Let $z_0 \in \mathbb{C} \wedge z_0 = \infty$.

$$\lim f(z) = \infty \iff \lim \frac{1}{f} = 0 \quad (8)$$

Proof. Suppose $z_0 = \infty$. Then,

$$\lim f(z) = \infty \iff \forall \epsilon > 0, \exists \delta > 0 : |z| > 1/\delta, z \in G \implies |f(z)| > 1/\epsilon \quad (9)$$

Then

$$\lim 1/f(z) = 0 \iff |z| > 1/\delta \implies 1/|f(z)| < \epsilon \quad (10)$$

Therefore, $\lim 1/f(z) = 0 \iff \lim f(z) = \infty$. $\square$

Example

Prove that:

$$\lim_{z \rightarrow \infty} \frac{z^2 + 1}{z - 1} = \infty \quad (11)$$

Proof. Use the useful fact.

$$\lim_{z \rightarrow \infty} \frac{z^2 + 1}{z - 1} = \infty \iff \lim_{z \rightarrow \infty} \frac{z - 1}{z^2 + 1} = 0$$

Use the algebraic properties of limits, then

$$\frac{z/z^2 - 1/z^2}{z^2/z^2 + 1/z^2} = \frac{1/z - 1/z^2}{1 + 1/z^2} \rightarrow 0 \quad (12)$$

$\square$

Continuity of functions of complex variable with values in $\overline{\mathbb{C}}$

Recall that $f : G \subseteq \overline{\mathbb{C}} \rightarrow \overline{\mathbb{C}}$ is called continuous if
(same definitions of continuity)

1. $z_0 \in G$

Example

Let $f(z) = 1/z$, $z \neq 0$. Extend to $\tilde{f}$, then

$$\tilde{f}(z) = \begin{cases} 1/z & z \neq 0, \neq \infty \\ \infty & z = 0 \\ 0 & z = \infty \end{cases} \quad (13)$$

This may be on the midterm.

We consider the following cases. We consider the first case. Clearly, by the algebraic properties of cont. limits, we have that $1/z$ is clearly cont.

Consider the second case. $\lim 1/\tilde{f} = \lim 1/f = 0$, then it is clear that $\lim z = 0$.